Substituting into the definition of the covariance matrix,

$$\begin{aligned}\sigma'_{ij} &= \frac{1}{2}\left\langle\left\{\hat{R}'_i - \bar{R}'_i, \hat{R}'_j - \bar{R}'_j\right\}\right\rangle \\ &= \frac{1}{2}\left\langle\left\{(S(\hat{R}-\bar{R}))_i, (S(\hat{R}-\bar{R}))_j\right\}\right\rangle.\end{aligned}$$

Using linearity,

$$(S(\hat{R}-\bar{R}))_i = \sum_k S_{ik}(\hat{R}_k - \bar{R}_k), \tag{2.37}$$

we obtain

$$\begin{aligned}\sigma'_{ij} &= \sum_{k,l} S_{ik}S_{jl}\frac{1}{2}\left\langle\left\{\hat{R}_k - \bar{R}_k, \hat{R}_l - \bar{R}_l\right\}\right\rangle \\ &= \sum_{k,l} S_{ik}\sigma_{kl}S_{jl}.\end{aligned}$$

Hence, in matrix form,

$$\sigma \to \sigma' = S\sigma S^T. \tag{2.38}$$

Under such transformations:

$$\sigma \to S\sigma S^T \tag{2.39}$$

Now, we see how **Gaussian states** are transformed:

The Wigner function for Gaussian states is:

$$W(R) = \frac{\exp\left[-\frac{1}{2}(R-\bar{R})^T\sigma^{-1}(R-\bar{R})\right]}{(2\pi)^N\sqrt{\det\sigma}} \tag{2.40}$$

With $R' = SR + d$, we invert to $R = S^{-1}(R' - d)$. The exponent transforms as:

$$\begin{aligned}(R-\bar{R})^T\sigma^{-1}(R-\bar{R}) &= (R'-\bar{R}')^T(S^{-1})^T\sigma^{-1}S^{-1}(R'-\bar{R}') \\ &= (R'-\bar{R}')^T(S\sigma S^T)^{-1}(R'-\bar{R}')\end{aligned}$$

where $\bar{R}' = S\bar{R} + d$ and $\sigma' = S\sigma S^T$. Since $\det S = 1$, $\det\sigma' = \det\sigma$. Thus:

$$W'(R') = \frac{\exp\left[-\frac{1}{2}(R'-\bar{R}')^T(\sigma')^{-1}(R'-\bar{R}')\right]}{(2\pi)^N\sqrt{\det\sigma'}} \tag{2.41}$$

Thus, Gaussian states are mapped to Gaussian states, and such unitary transformations are also called **Gaussian transformations**.

2.7 Characteristic Functions (Quantum)

The characteristic function of a quantum state ρ is defined as

$$\chi(\alpha) = \mathrm{Tr}\left[\rho D(\alpha)\right], \quad D(\alpha) = \exp\left(\alpha a^\dagger - \alpha^* a\right) \tag{2.42}$$

(for displacement operator $D(\alpha)$ refer to eq. (2.64)) Because a and $a^\dagger$ do not commute, different operator orderings give different characteristic functions [2], which correspond to different quasi-probability distributions in phase space::

$$\chi(\alpha) = \mathrm{Tr}[\rho D(\alpha)] \quad \text{(Weyl / symmetric)}, \tag{2.43}$$

$$\chi_N(\alpha) = \mathrm{Tr}\left[\rho e^{\alpha a^\dagger} e^{-\alpha^* a}\right] \quad \text{(normal ordering)}, \tag{2.44}$$

$$\chi_A(\alpha) = \mathrm{Tr}\left[\rho e^{-\alpha^* a} e^{\alpha a^\dagger}\right] \quad \text{(anti-normal ordering)}. \tag{2.45}$$

These correspond respectively to different phase-space distributions (Wigner, P, and Q functions).

For Gaussian states,$\chi(\xi) = \exp\left(-\frac{1}{2}\xi^T \sigma \xi + i\,\xi^T \Omega \bar{R}\right)$

Here $\xi \in \mathbb{R}^{2n}$ denotes the real phase-space vector corresponding to the complex displacement α. So, the state is fully determined by mean $\bar{R}$ and covariance σ.

Relation to Classical Probability:

In classical theory, the characteristic function of a random variable x is

$$\chi_{ct}(\lambda) = \mathbb{E}[e^{i\lambda x}] = \int p(x) e^{i\lambda x} dx. \tag{2.46}$$

The quantum version is a direct analogue: as if we are finding expectation value ($\mathrm{Tr}[\rho D(\alpha)]$) of $D(\alpha)$ (like of $e^{i\lambda x}$ in classical theory).

2.8 Wigner Function

The Wigner function provides a phase-space representation of a quantum state [3]. For a single-mode density operator $\hat{\rho}$, it is defined as

$$W(q,p) = \frac{1}{\pi\hbar} \int_{-\infty}^{\infty} dy\, \langle q - y|\hat{\rho}|q + y\rangle\, e^{2ipy/\hbar} \tag{2.47}$$

Characteristic Function Representation: The Wigner function provides an equivalent description of an n-mode quantum state in phase space to the characteristic function. It can be defined as the Fourier transform of the symmetrically ordered characteristic function $\chi(\boldsymbol{\lambda})$:

$$W(\boldsymbol{\alpha}) = \int \frac{d^{2n}\boldsymbol{\lambda}}{\pi^{2n}} \chi(\boldsymbol{\lambda}) \exp\big(\boldsymbol{\alpha}\boldsymbol{\lambda}^\dagger - \boldsymbol{\alpha}^\dagger\boldsymbol{\lambda}\big) \tag{2.48}$$

For a single mode, this reduces to

$$W(\alpha) = \int \frac{d^{2}\boldsymbol{\lambda}}{\pi^{2}} \chi(\boldsymbol{\lambda}) \exp\big(\boldsymbol{\alpha}\boldsymbol{\lambda}^\dagger - \boldsymbol{\alpha}^\dagger\boldsymbol{\lambda}\big) \tag{2.49}$$

Here, $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_n)^T$ is the vector of complex phase-space coordinates, where each component is related to the quadrature operators by $\alpha_k = (q_k + ip_k)/\sqrt{2}$. The variable $\boldsymbol{\lambda}$ is the integration variable in the Fourier transform, corresponding to the argument of the displacement operator in the characteristic function $\chi(\boldsymbol{\lambda}) = \mathrm{Tr}[\hat{\rho}\hat{D}(\boldsymbol{\lambda})]$.

Properties:

- Normalization: $\int dq\, dp\, W(q,p) = 1$.
- $W(q,p)$ is real, but not necessarily positive.
- Marginals:
$$\int dp\, W(q,p) = \langle q|\hat{\rho}|q\rangle, \qquad \int dq\, W(q,p) = \langle p|\hat{\rho}|p\rangle.$$

Expectation Values:

$$\langle \hat{A} \rangle = \int d^{2n}\mathbf{R}\, W_\rho(\mathbf{R})\, A_W(\mathbf{R}), \tag{2.50}$$

where $A_W(\mathbf{R})$ is the Weyl symbol of $\hat{A}$.

2.9 P Representation

The Glauber–Sudarshan P-distribution provides a diagonal representation of the density operator in the coherent-state basis:

$$\hat{\rho} = \int d^2\alpha\, P(\alpha)\, |\alpha\rangle\langle\alpha|. \tag{2.51}$$

Properties:

- **Diagonal form:** Expresses $\hat{\rho}$ as a weighted mixture of coherent states.
- **Classicality:** If $P(\alpha)$ behaves like a true probability distribution (positive and regular), the state is classical.
- **Non-classicality:** For many quantum states (e.g., Fock states), $P(\alpha)$ becomes highly singular or non-positive.

- **Characteristic Function:** Related to the normally ordered characteristic function:

$$\chi_N(\lambda) = \int d^2\alpha\, P(\alpha)\, e^{\lambda\alpha^* - \lambda^*\alpha}. \tag{2.52}$$

2.10 Trace Rules in Phase Space

Trace identities allow operator expressions to be evaluated as integrals in phase space, which are often simpler to compute.

Glauber Expansion: Any operator $\hat{O}$ acting on an n-mode Hilbert space can be written as

$$\hat{O} = \int \frac{d^{2n}\lambda}{\pi^n}\, \chi[\hat{O}](\lambda)\, \hat{D}(-\lambda), \tag{2.53}$$

where $\chi[\hat{O}](\lambda)$ is the characteristic function.

Trace Rule (Characteristic Functions): For two operators $\hat{O}_1$ and $\hat{O}_2$:

$$\mathrm{Tr}(\hat{O}_1\hat{O}_2) = \int \frac{d^{2n}\lambda}{\pi^n}\, \chi[\hat{O}_1](\lambda)\, \chi[\hat{O}_2](-\lambda). \tag{2.54}$$

Trace Rule (Wigner Functions): Equivalently, in terms of Wigner functions:

$$\mathrm{Tr}(\hat{O}_1\hat{O}_2) = (\pi)^n \int d^{2n}\lambda\, W[\hat{O}_1](\lambda)\, W[\hat{O}_2](\lambda). \tag{2.55}$$

which will be used to derive the fidelity and Quantum Chernoff Bound for Gaussian states, in later sections.

2.11 Uncertainty Principle

The uncertainty principle follows from the non-commutativity of operators. For any two observables A and B, the Robertson uncertainty relation is given by

$$\Delta A \Delta B \geq \frac{1}{2}\, |\langle [A, B] \rangle| \tag{2.56}$$

For continuous variable systems, the canonical quadrature operators $\hat{q}$ and $\hat{p}$ satisfy

$$[\hat{q}, \hat{p}] = i\hbar, \tag{2.57}$$

which leads to the standard uncertainty relation

$$\Delta q \Delta p \geq \frac{\hbar}{2} \tag{2.58}$$

More generally, for an n-mode system described by the covariance matrix σ, the uncertainty principle can be written in compact form [4] as

$$\sigma + \frac{i\hbar}{2}\Omega \geq 0 \tag{2.59}$$

where Ω is the symplectic form.

2.12 Coherent States

2.12.1 Fock representation of the coherent state

Assume that the annihilation operator $\hat{a}$ has a right eigenstate. Because $\hat{a}$ is not Hermitian, its eigenvalue will in general be some complex number α, and we label the corresponding eigenstate $|\alpha\rangle$ by the same complex number α. We may therefore write the equation

$$\hat{a}|\alpha\rangle = \alpha|\alpha\rangle.$$

The conjugate state $\langle\alpha|$ is obviously a left eigenstate of the creation operator $\hat{a}^\dagger$, as can be seen by taking the Hermitian conjugate.

$$\langle\alpha|\hat{a}^\dagger = \alpha^*\langle\alpha|.$$

The state $|\alpha\rangle$ will be called the coherent state. Our initial goal is to find an explicit representation for $|\alpha\rangle$ in the basis of Fock states $|n\rangle$ labelled by the occupation number n. Since the Fock states form a complete set, we may use them to represent $|\alpha\rangle$ in the form

$$|\alpha\rangle = \sum_{n=0}^{\infty} c_n|n\rangle,$$

in which the c_n are complex numbers to be determined. On substituting the expansion and using the "step down" equation, we obtain

$$\sum_{n=1}^{\infty} c_n\sqrt{n}|n-1\rangle = \alpha\sum_{n=0}^{\infty} c_n|n\rangle.$$

As the $|n\rangle$ $(n = 0, 1, 2, ...)$ are a set of orthogonal state vectors, this equation is satisfied only if the coefficients of corresponding Fock state vectors on both sides are equal. We therefore have, on equating coefficients of $|n-1\rangle$,

$$c_n = \frac{\alpha}{\sqrt{n}}c_{n-1}.$$

This is a recursion formula connecting c_n and c_{n-1} from which we obtain by repeated

application

$$c_n = \frac{\alpha^2}{[n(n-1)]^{1/2}} c_{n-2} = ... = \frac{\alpha^n}{\sqrt{n!}} c_0,$$

so that

$$|\alpha\rangle = c_0 \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle.$$

c_0 can be determined from the requirement that the state $|\alpha\rangle$ be normalized to unity, which implies

$$\begin{aligned}\langle\alpha|\alpha\rangle = 1 = |c_0|^2 \sum_{n=0}^{\infty}\sum_{m=0}^{\infty} \frac{\alpha^{*n}\alpha^m}{\sqrt{n!}\sqrt{m!}} \langle n|m\rangle \\ = |c_0|^2 \sum_{n=0}^{\infty} \frac{|\alpha|^{2n}}{n!} \\ = |c_0|^2 e^{|\alpha|^2},\end{aligned}$$

when we make use of the orthogonality of the Fock states. Hence

$$|c_0| = e^{-|\alpha|^2/2},$$

and, apart from a unimodular factor, we have

$$|\alpha\rangle = e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle, \tag{2.60}$$

and

$$\langle\alpha| = e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^{*n}}{\sqrt{n!}} \langle n|.$$

It is interesting to note that for every complex number α, other than zero, the coherent state $|\alpha\rangle$ has a non-zero projection on every Fock state $|n\rangle$. Thus

$$\langle n|\alpha\rangle = e^{-|\alpha|^2/2} \frac{\alpha^n}{\sqrt{n!}}.$$

When $\alpha = 0$, the coherent state $|\alpha\rangle$ becomes the vacuum state $|vac\rangle$, which may be regarded as either a coherent state or a Fock state. The squared modulus of the projection of $|\alpha\rangle$ onto $|n\rangle$ gives the probability $p(n)$ that n excitations or photons will be found in the coherent state $|\alpha\rangle$. We therefore have

$$p(n) = |\langle n|\alpha\rangle|^2 = \frac{|\alpha|^{2n}}{n!} e^{-|\alpha|^2}, \tag{2.61}$$

which will be recognized as a Poisson distribution in n, with parameter $|\alpha|^2$. The mean number of photons present when the state is a coherent state $|\alpha\rangle$ is therefore given

by(Poisson distbn - mean is the λ of the distbn. - here $|\alpha|^2$).

$$\sum_{n=0}^{\infty} np(n) = |\alpha|^2 = \langle\alpha|\hat{a}^\dagger\hat{a}|\alpha\rangle \tag{2.62}$$

and this is large or small according as α is a large or small number.

But, no matter how small α may be (with the exception of $\alpha = 0$), there is some non-zero probability $p(n)$ that any number of photons n, however large, is present in the field. In a certain sense the photon number is as random as possible in a coherent state, subject to a certain mean number $|\alpha|^2$. It is a remarkable feature of the combination of Fock states that the state is unchanged when acted on by the annihilation operator $\hat{a}$. A possible connection between the coherent state of the quantum field and a classical field is suggested by the fact that it is possible to absorb photons from an electromagnetic field in a coherent state repeatedly, without changing the state in any way.

At first sight it might seem that, because the coherent state is not the eigenstate of any observable, it does not correspond to any readily measurable feature of the electromagnetic field. However, this is not so, because in practice most measurements of the field in the optical domain are based on the process of photoelectric detection. This includes instruments such as the photomultiplier, the photoconductor, the photographic plate and the eye. These instruments function by the absorption of photons, and it is therefore the absorption operator which is most closely associated with the measurement of the field. It is partly because the coherent states are eigenstates of the absorption operator, that these states prove to be particularly convenient for the description of many properties of the field encountered in photoelectric measurements.[5, Chapter 11]

2.12.2 The coherent state as a displaced vacuum state - the displacement operator

By combining the expansion with the representation of the Fock state $|n\rangle = \frac{\hat{a}^{\dagger n}}{\sqrt{n!}}|vac\rangle$, we can write

$$\begin{aligned}|\alpha\rangle &= e^{-|\alpha|^2/2}\sum_{n=0}^{\infty}\frac{\alpha^n\hat{a}^{\dagger n}}{n!}|vac\rangle \\ &= e^{-|\alpha|^2/2}e^{\alpha\hat{a}^\dagger}|vac\rangle,\end{aligned}$$

which shows that the coherent state $|\alpha\rangle$ may also be regarded as a displaced vacuum state .

We can re-express this in a somewhat more symmetric form, by inserting the operator $e^{-\alpha^*\hat{a}}$ between $e^{\alpha\hat{a}^\dagger}$ and $|vac\rangle$. Thus

$$|\alpha\rangle = e^{-|\alpha|^2/2}e^{\alpha\hat{a}^\dagger}e^{-\alpha^*\hat{a}}|vac\rangle. \tag{2.63}$$

The insertion of $e^{-\alpha^*\hat{a}}$ can be justified by expansion of the exponential operator, which leads to the result

$$\begin{aligned} e^{-\alpha^*\hat{a}}|vac\rangle &= \left[1 - \alpha^*\hat{a} + \frac{(\alpha^*\hat{a})^2}{2!} - ...\right]|vac\rangle \\ &= |vac\rangle. \end{aligned}$$

We now make use of the Campbell-Baker-Hausdorff operator identity for two operators $\hat{A}$, $\hat{B}$ provided that

$$\exp\Big(\hat{A} + \hat{B}\Big) = \exp\hat{A}\exp\hat{B}\exp\Big(-[\hat{A}, \hat{B}]/2\Big),$$

$$[\hat{A}, [\hat{A}, \hat{B}]] = 0 = [\hat{B}, [\hat{A}, \hat{B}]].$$

The condition is obviously satisfied for any pair of operators $\hat{A}$, $\hat{B}$ whose commutator $[\hat{A}, \hat{B}]$ is a c-number. If we now put $\hat{A} = \alpha\hat{a}^\dagger$, $\hat{B} = -\alpha^*\hat{a}$ and make use of the commutation relation, we obtain $[\hat{A}, \hat{B}] = |\alpha|^2$ and

$$e^{\alpha\hat{a}^\dagger - \alpha^*\hat{a}} = e^{\alpha\hat{a}^\dagger}e^{-\alpha^*\hat{a}}e^{-|\alpha|^2/2}$$

which allows us to re-write the equation in the more compact form

$$|\alpha\rangle = \hat{D}(\alpha)|0\rangle, \tag{2.64}$$

where

$$\hat{D}(\alpha) \equiv e^{\alpha\hat{a}^\dagger - \alpha^*\hat{a}}.$$

$\hat{D}(\alpha)$ is the displacement operator that creates the coherent state $|\alpha\rangle$ from the vacuum state $|0\rangle = |vac\rangle$. It is clear by inspection that $\hat{D}(\alpha)$ is a unitary operator such that

$$\hat{D}^\dagger(\alpha)\hat{D}(\alpha) = 1 = \hat{D}(\alpha)\hat{D}^\dagger(\alpha),$$

and

$$\hat{D}^\dagger(\alpha) = \hat{D}(-\alpha).$$

The displacement operator also has the following property:

Unitary transformation of $\hat{a}$ or $\hat{a}^\dagger$ leads to the complex displacement α or α^*, respectively. Thus

$$\hat{D}^\dagger(\alpha)\hat{a}\hat{D}(\alpha) = \hat{a} + \alpha$$

and

$$\hat{D}^\dagger(\alpha)\hat{a}^\dagger\hat{D}(\alpha) = \hat{a}^\dagger + \alpha^*.$$

These properties follow by applying the operator expansion theorem (the Hadamard

lemma), which states that for any operators $\hat{A}$ and $\hat{B}$:

$$e^{\hat{A}}\hat{B}e^{-\hat{A}} = \hat{B} + [\hat{A}, \hat{B}] + \frac{1}{2!}[\hat{A}, [\hat{A}, \hat{B}]] + \dots$$

Letting the displacement operator be $\hat{D}(\alpha) = e^{\alpha\hat{a}^\dagger - \alpha^*\hat{a}}$, its adjoint is $\hat{D}^\dagger(\alpha) = e^{-\alpha\hat{a}^\dagger + \alpha^*\hat{a}}$. To map this to the theorem, we set $\hat{A} = -\alpha\hat{a}^\dagger + \alpha^*\hat{a}$. First, we evaluate the transformation of $\hat{B} = \hat{a}$. Using the fundamental commutation relations $[\hat{a}, \hat{a}^\dagger] = 1$ and $[\hat{a}^\dagger, \hat{a}] = -1$, we compute the first-order commutator:

$$\begin{aligned}
[\hat{A}, \hat{a}] &= [-\alpha\hat{a}^\dagger + \alpha^*\hat{a}, \hat{a}] \\
&= -\alpha[\hat{a}^\dagger, \hat{a}] + \alpha^*[\hat{a}, \hat{a}] \\
&= -\alpha(-1) + 0 \\
&= \alpha
\end{aligned}$$

Because α is a c-number (scalar), it commutes with all operators. Therefore, all higher-order nested commutators strictly vanish:

$$[\hat{A}, [\hat{A}, \hat{a}]] = [\hat{A}, \alpha] = 0$$

Substituting these terms back into the expansion theorem, the infinite series truncates exactly after the second term:

$$\hat{D}^\dagger(\alpha)\hat{a}\hat{D}(\alpha) = e^{-\alpha\hat{a}^\dagger + \alpha^*\hat{a}}\hat{a}e^{\alpha\hat{a}^\dagger - \alpha^*\hat{a}} = \hat{a} + \alpha$$

We repeat the process for $\hat{B} = \hat{a}^\dagger$. The first-order commutator evaluates to:

$$\begin{aligned}
[\hat{A}, \hat{a}^\dagger] &= [-\alpha\hat{a}^\dagger + \alpha^*\hat{a}, \hat{a}^\dagger] \\
&= -\alpha[\hat{a}^\dagger, \hat{a}^\dagger] + \alpha^*[\hat{a}, \hat{a}^\dagger] \\
&= 0 + \alpha^*(1) \\
&= \alpha^*
\end{aligned}$$

Once again, the resulting c-number α^* causes all higher-order nested commutators ($[\hat{A}, \alpha^*]$) to vanish. The expansion truncates cleanly, yielding:

$$\hat{D}^\dagger(\alpha)\hat{a}^\dagger\hat{D}(\alpha) = e^{-\alpha\hat{a}^\dagger + \alpha^*\hat{a}}\hat{a}^\dagger e^{\alpha\hat{a}^\dagger - \alpha^*\hat{a}} = \hat{a}^\dagger + \alpha^*$$

[5, Chapter 11]

2.13 Thermal States

2.13.1 The Density Matrix

We begin with the fundamental postulate of statistical mechanics for a system in thermal equilibrium with a heat bath. The canonical ensemble defines the unnormalized density operator as $e^{-\beta\hat{H}}$, where $\beta = \frac{1}{k_B T}$ and $\hat{H}$ is the Hamiltonian. For a quantum harmonic oscillator, the Hamiltonian is $\hat{H} = \hbar\omega\left(\hat{a}^\dagger\hat{a} + \frac{1}{2}\right)$.

To find the normalized density operator $\hat{\rho}_{th}$, we divide by the partition function Z:

$$\hat{\rho}_{th} = \frac{1}{Z} e^{-\beta\hbar\omega\left(\hat{a}^\dagger\hat{a}+1/2\right)} \tag{2.65}$$

We evaluate the partition function Z by taking the trace over the Fock state basis $|n\rangle$:

$$\begin{aligned} Z &= \mathrm{Tr}\left(e^{-\beta\hat{H}}\right) \\ &= \sum_{n=0}^{\infty} \langle n| e^{-\beta\hbar\omega\left(\hat{a}^\dagger\hat{a}+1/2\right)} |n\rangle \\ &= e^{-\frac{\beta\hbar\omega}{2}} \sum_{n=0}^{\infty} e^{-\beta\hbar\omega n} \end{aligned}$$

Because $e^{-\beta\hbar\omega} < 1$, we recognize this as a convergent geometric series $\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$. Evaluating the series yields:

$$Z = e^{-\frac{\beta\hbar\omega}{2}} \left(\frac{1}{1 - e^{-\beta\hbar\omega}}\right)$$

Substituting Z back into the density operator cancels the zero-point energy factor $e^{-\frac{\beta\hbar\omega}{2}}$:

$$\hat{\rho}_{th} = \left(1 - e^{-\beta\hbar\omega}\right) \sum_{n=0}^{\infty} e^{-\beta\hbar\omega n} |n\rangle\langle n|$$

We now translate this thermodynamic expression into the language of continuous variables by parameterizing it with the average photon number, $N = \langle\hat{a}^\dagger\hat{a}\rangle$. Applying Bose-Einstein statistics gives:

$$N = \frac{1}{e^{\beta\hbar\omega} - 1}$$

Rearranging this relation to solve for $e^{-\beta\hbar\omega}$ gives:

$$\begin{aligned} e^{\beta\hbar\omega} &= \frac{N+1}{N} \\ e^{-\beta\hbar\omega} &= \frac{N}{N+1} \end{aligned}$$

We also determine the normalization coefficient:

$$1 - e^{-\beta\hbar\omega} = 1 - \frac{N}{N+1} = \frac{1}{N+1}$$

Substituting these expressions back into the density operator yields the final density matrix in the Fock basis:

$$\hat{\rho}_{th} = \frac{1}{N+1} \sum_{n=0}^{\infty} \left(\frac{N}{N+1} \right)^n |n\rangle\langle n| \tag{2.66}$$

2.13.2 The Mean Vector

The mean vector $\bar{R}$ consists of the expectation values of the quadrature operators, $\bar{R} = (\langle\hat{q}\rangle, \langle\hat{p}\rangle)^T$. We define the quadrature operators in terms of the creation and annihilation operators:

$$\hat{q} = \frac{1}{2}(\hat{a} + \hat{a}^\dagger), \quad \hat{p} = \frac{1}{2i}(\hat{a} - \hat{a}^\dagger)$$

To compute $\langle\hat{q}\rangle$, we first evaluate the expectation values of $\hat{a}$ and $\hat{a}^\dagger$:

$$\langle\hat{a}\rangle = \mathrm{Tr}(\hat{\rho}_{th}\hat{a}) = \sum_{n=0}^{\infty} \frac{1}{N+1} \left(\frac{N}{N+1} \right)^n \langle n|\hat{a}|n\rangle$$

Applying the annihilation operator yields $\hat{a}|n\rangle = \sqrt{n}|n-1\rangle$. Because Fock states are orthogonal, the inner product $\langle n|n-1\rangle$ equals 0. Consequently, $\langle\hat{a}\rangle = 0$. Identical logic proves $\langle\hat{a}^\dagger\rangle = 0$.

Because both operator expectations vanish, the quadrature expectations also vanish:

$$\langle\hat{q}\rangle = 0, \quad \langle\hat{p}\rangle = 0$$

This gives the final mean vector:

$$\bar{R} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} = 0 \tag{2.67}$$

2.13.3 The Covariance Matrix

The elements of the covariance matrix σ follow the definition:

$$\sigma_{ij} = \frac{1}{2}\langle\{\hat{R}_i, \hat{R}_j\}\rangle - \langle\hat{R}_i\rangle\langle\hat{R}_j\rangle$$

Because the mean vector $\bar{R}$ equals 0, the second term drops out, leaving only the symmetric anti-commutator expectations.

Position Variance (σ_{11})

We calculate $\langle \hat{q}^2 \rangle$ by expanding the square of the position quadrature:

$$\hat{q}^2 = \frac{1}{4}\left(\hat{a} + \hat{a}^\dagger\right)^2 = \frac{1}{4}\left(\hat{a}^2 + (\hat{a}^\dagger)^2 + \hat{a}\hat{a}^\dagger + \hat{a}^\dagger\hat{a}\right)$$

Using the canonical commutation relation $[\hat{a}, \hat{a}^\dagger] = 1$, we rewrite $\hat{a}\hat{a}^\dagger = \hat{a}^\dagger\hat{a} + 1$. This transforms the expression to:

$$\hat{q}^2 = \frac{1}{4}\left(\hat{a}^2 + (\hat{a}^\dagger)^2 + 2\hat{a}^\dagger\hat{a} + 1\right)$$

Taking the expectation value, the diagonal nature of the thermal state ensures that $\langle \hat{a}^2 \rangle = 0$ and $\langle (\hat{a}^\dagger)^2 \rangle = 0$. We substitute the known expectation $\langle \hat{a}^\dagger\hat{a} \rangle = N$ to find:

$$\sigma_{11} = \langle \hat{q}^2 \rangle = \frac{1}{4}(2N + 1)$$

Momentum Variance (σ_{22})

We apply the same process to calculate $\langle \hat{p}^2 \rangle$:

$$\begin{aligned}
\hat{p}^2 &= -\frac{1}{4}\left(\hat{a} - \hat{a}^\dagger\right)^2 \\
&= -\frac{1}{4}\left(\hat{a}^2 + (\hat{a}^\dagger)^2 - \hat{a}\hat{a}^\dagger - \hat{a}^\dagger\hat{a}\right) \\
&= -\frac{1}{4}\left(\hat{a}^2 + (\hat{a}^\dagger)^2 - 2\hat{a}^\dagger\hat{a} - 1\right)
\end{aligned}$$

Taking the expectation value and eliminating the vanishing terms yields:

$$\sigma_{22} = \langle \hat{p}^2 \rangle = -\frac{1}{4}(-2N - 1) = \frac{1}{4}(2N + 1)$$

Cross-Terms (σ_{12} and σ_{21})

We evaluate the anti-commutator $\frac{1}{2}\langle \hat{q}\hat{p} + \hat{p}\hat{q} \rangle$:

$$\begin{aligned}
\hat{q}\hat{p} + \hat{p}\hat{q} &= \frac{1}{4i}\left[(\hat{a} + \hat{a}^\dagger)(\hat{a} - \hat{a}^\dagger) + (\hat{a} - \hat{a}^\dagger)(\hat{a} + \hat{a}^\dagger)\right] \\
&= \frac{1}{4i}\left[(\hat{a}^2 - \hat{a}\hat{a}^\dagger + \hat{a}^\dagger\hat{a} - (\hat{a}^\dagger)^2) + (\hat{a}^2 + \hat{a}\hat{a}^\dagger - \hat{a}^\dagger\hat{a} - (\hat{a}^\dagger)^2)\right] \\
&= \frac{1}{4i}\left[2\hat{a}^2 - 2(\hat{a}^\dagger)^2\right] \\
&= \frac{1}{2i}\left(\hat{a}^2 - (\hat{a}^\dagger)^2\right)
\end{aligned}$$

Because the expectation values of $\hat{a}^2$ and $(\hat{a}^\dagger)^2$ are both zero, the cross-terms vanish: $\sigma_{12} = \sigma_{21} = 0$.

Assembling these components provides the complete covariance matrix:

$$\sigma = \frac{1}{4}\begin{pmatrix} 2N+1 & 0 \\ 0 & 2N+1 \end{pmatrix} \tag{2.68}$$

2.13.4 The Wigner Function

Because the thermal state is a Gaussian state, we can construct its Wigner function entirely from its covariance matrix and mean vector. The standard mathematical definition of the Wigner function for a zero-mean Gaussian state is:

$$W(\mathbf{r}) = \frac{1}{2\pi\sqrt{\det\sigma}} \exp\left(-\frac{1}{2}\mathbf{r}^T\sigma^{-1}\mathbf{r}\right)$$

where $\mathbf{r} = (x, p)^T$.

First, we calculate the determinant of the covariance matrix:

$$\det\sigma = \left(\frac{2N+1}{4}\right)^2$$

Taking the square root yields $\sqrt{\det\sigma} = \frac{2N+1}{4}$.

We use this to establish the normalization coefficient at the front of the Wigner function:

$$\frac{1}{2\pi\sqrt{\det\sigma}} = \frac{1}{2\pi\left(\frac{2N+1}{4}\right)} = \frac{2}{\pi(2N+1)}$$

Next, we invert the diagonal covariance matrix:

$$\sigma^{-1} = \frac{4}{2N+1}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Finally, we compute the matrix multiplication inside the exponent:

$$\begin{aligned} -\frac{1}{2}\mathbf{r}^T\sigma^{-1}\mathbf{r} &= -\frac{1}{2}\begin{pmatrix} x & p \end{pmatrix}\left[\frac{4}{2N+1}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\right]\begin{pmatrix} x \\ p \end{pmatrix} \\ &= -\frac{2}{2N+1}(x^2+p^2) \end{aligned}$$

Substituting the normalization coefficient and the expanded exponent into the general Gaussian formula produces the corrected Wigner function for the thermal state:

$$W(x,p) = \frac{2}{\pi(2N+1)} \exp\left(-\frac{2}{2N+1}(x^2+p^2)\right) \tag{2.69}$$

2.14 Squeeze Operator

2.14.1 The unitary squeeze operator

It is possible to generate a squeezed single-mode state from an unsqueezed one by the action of the following simple unitary operator,

$$\hat{S}(\xi) = \exp \frac{1}{2}(\xi^* \hat{a}^2 - \xi \hat{a}^{\dagger 2}), \quad \xi = re^{i\theta},$$

which is known as the squeeze operator.

To find the mode evolution of the annihilation operator $\hat{a}$ under the single-mode squeezing operator $\hat{S}(\xi)$, we apply the Baker-Campbell-Hausdorff (BCH) Lemma.

2.14.2 Defining the Operators

The squeezing operator for a single boson is defined using the squeezing parameter ξ:

$$\hat{S}(\xi) = \exp\left(\frac{1}{2}\xi^* \hat{a}^2 - \frac{1}{2}\xi(\hat{a}^\dagger)^2\right) \tag{2.70}$$

This squeezing parameter ξ is complex and can be expressed as $\xi = re^{i\theta}$, a value determined by the squeezing amplitude r and the squeezing angle θ.

Let the exponent be the operator $\hat{G}$, meaning $\hat{S}(\xi) = \exp\left(\hat{G}\right)$:

$$\hat{G} = \frac{1}{2}\xi^* \hat{a}^2 - \frac{1}{2}\xi(\hat{a}^\dagger)^2$$

Because $\hat{G}$ is an anti-Hermitian operator ($\hat{G}^\dagger = -\hat{G}$), the adjoint of the squeezing operator is $\hat{S}(\xi)^\dagger = \exp\left(-\hat{G}\right)$. The full transformation we need to calculate is therefore:

$$\hat{S}(\xi)^\dagger \hat{a} \hat{S}(\xi) = \exp\left(-\hat{G}\right) \hat{a} \exp\left(\hat{G}\right)$$

2.14.3 Applying the Baker-Campbell-Hausdorff Lemma

For generic operators $\hat{A}$ and $\hat{B}$, the BCH lemma states:

$$e^{\hat{A}} \hat{B} e^{-\hat{A}} = \hat{B} + [\hat{A}, \hat{B}] + \frac{1}{2!}[\hat{A}, [\hat{A}, \hat{B}]] + \frac{1}{3!}[\hat{A}, [\hat{A}, [\hat{A}, \hat{B}]]] + \dots$$

Substituting $\hat{A} = -\hat{G}$ and $\hat{B} = \hat{a}$, the formula generates an alternating series of nested commutators:

$$e^{-\hat{G}} \hat{a} e^{\hat{G}} = \hat{a} - [\hat{G}, \hat{a}] + \frac{1}{2!}[\hat{G}, [\hat{G}, \hat{a}]] - \frac{1}{3!}[\hat{G}, [\hat{G}, [\hat{G}, \hat{a}]]] + \dots$$

2.14.4 Calculating the Nested Commutators

With $[\hat{a}, \hat{a}^\dagger] = 1$, the terms evaluate as follows.

First Commutator:

$$\begin{aligned}[\hat{G}, \hat{a}] &= \left[\frac{1}{2}\xi^*\hat{a}^2 - \frac{1}{2}\xi(\hat{a}^\dagger)^2, \hat{a}\right] \\ &= -\frac{1}{2}\xi[(\hat{a}^\dagger)^2, \hat{a}] + \frac{1}{2}\xi^*[\hat{a}^2, \hat{a}]\end{aligned}$$

Since $[\hat{a}^2, \hat{a}] = 0$, the remaining term expands via the product identity $[\hat{A}\hat{B}, \hat{C}] = \hat{A}[\hat{B}, \hat{C}] + [\hat{A}, \hat{C}]\hat{B}$:

$$\begin{aligned}[(\hat{a}^\dagger)^2, \hat{a}] &= \hat{a}^\dagger[\hat{a}^\dagger, \hat{a}] + [\hat{a}^\dagger, \hat{a}]\hat{a}^\dagger \\ &= \hat{a}^\dagger(-1) + (-1)\hat{a}^\dagger = -2\hat{a}^\dagger\end{aligned}$$

Therefore:

$$[\hat{G}, \hat{a}] = -\frac{1}{2}\xi(-2\hat{a}^\dagger) = \xi\hat{a}^\dagger$$

Second Commutator:

$$\begin{aligned}[\hat{G}, [\hat{G}, \hat{a}]] &= [\hat{G}, \xi\hat{a}^\dagger] = \xi[\hat{G}, \hat{a}^\dagger] \\ [\hat{G}, \hat{a}^\dagger] &= \left[\frac{1}{2}\xi^*\hat{a}^2 - \frac{1}{2}\xi(\hat{a}^\dagger)^2, \hat{a}^\dagger\right] = \frac{1}{2}\xi^*[\hat{a}^2, \hat{a}^\dagger]\end{aligned}$$

The term $[\hat{a}^2, \hat{a}^\dagger]$ evaluates to:

$$[\hat{a}^2, \hat{a}^\dagger] = \hat{a}[\hat{a}, \hat{a}^\dagger] + [\hat{a}, \hat{a}^\dagger]\hat{a} = \hat{a}(1) + (1)\hat{a} = 2\hat{a}$$

Consequently:

$$[\hat{G}, \hat{a}^\dagger] = \frac{1}{2}\xi^*(2\hat{a}) = \xi^*\hat{a}$$

and the nested commutator becomes:

$$[\hat{G}, [\hat{G}, \hat{a}]] = \xi(\xi^*\hat{a}) = |\xi|^2\hat{a} = r^2\hat{a}$$

Higher-Order Commutators: Subsequent commutators scale by factors of r^2:

$$\begin{aligned}[\hat{G}, [\hat{G}, [\hat{G}, \hat{a}]]] &= [\hat{G}, r^2\hat{a}] = r^2[\hat{G}, \hat{a}] = r^2\xi\hat{a}^\dagger \\ [\hat{G}, [\hat{G}, [\hat{G}, [\hat{G}, \hat{a}]]]] &= [\hat{G}, r^2\xi\hat{a}^\dagger] = r^2\xi[\hat{G}, \hat{a}^\dagger] = r^4\hat{a}\end{aligned}$$